Contents

Q1 First Order

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-5,5]; % interval
points=200; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
U=zeros(points,1)';P=zeros(points,1)';
U(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi)); % intial conditions
P(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi));
Ui=U;
Pi=P;
tim=10;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
matA=[-1 1;1 1];
AU=[];BU=[];AP=[];BP=[];
for i=1:n
    % A be forward difference and B to be backward
    AU(1:points-1)=U(2:points)-U(1:points-1);
    AU(points)=-U(points-1);
    BU(2:points)=U(2:points)-U(1:points-1);
    BU(1)=U(1);

    AP(1:points-1)=P(2:points)-P(1:points-1);
    AP(points)=0;
    BP(2:points)=P(2:points)-P(1:points-1);
    BP(1)=0;

    coffpos=inv(matA)*[AP;AU];
    coffnec=inv(matA)*[BP;BU];


    P=P-dtdx*(coffnec(2,:) + coffpos(1,:));
    U=U-dtdx*(coffnec(2,:) - coffpos(1,:));

end
```

Q1 second Order

```
clc
clear
dtdx=0.8; % dt/dx
```

```
inter=[-5,5]; % interval
points=200; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
U=zeros(points,1)';P=zeros(points,1)';
U(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi)); % intial conditions
P(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi));
Ui=U;
Pi=P;
tim=10;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
matA=[-1 1;1 1];
for i=1:n
    % A be forward difference and B to be backward
    AU(1:points-1)=U(2:points)-U(1:points-1);
    AU(points)=-U(points-1);
    BU(2:points)=U(2:points)-U(1:points-1);
    BU(1)=U(1);

    AP(1:points-1)=P(2:points)-P(1:points-1);
    AP(points)=0;
    BP(2:points)=P(2:points)-P(1:points-1);
    BP(1)=0;

    coffpos=inv(matA)*[AP;AU];
    coffnec=inv(matA)*[BP;BU];

    mupos=(1+dtdx)/2;
    munev=(1-dtdx)/2;

    PFnev=mupos*coffnec(2,:) + munev*coffnec(1,:);
    PFpos=munev*coffpos(2,:) + mupos*coffpos(1,:);

    UFnev=mupos*coffnec(2,:) - munev*coffnec(1,:);
    UFpos=munev*coffpos(2,:) - mupos*coffpos(1,:);

    P=P-dtdx*(PFnev + PFpos);
    U=U-dtdx*(UFnev + UFpos);

end
```

Q2 First Order @t=6

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-5,5]; % interval
points=200; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
U=zeros(points,1)';P=zeros(points,1)';
U(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi)); % intial conditions
P(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi));
Ui=U;
Pi=P;
tim=6;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
matA=[-1 1;1 1];
```

```
AU=[];BU=[];AP=[];BP=[];
for i=1:n
    % A be forward difference and B to be backward
    AU(1:points-1)=U(2:points)-U(1:points-1);
    AU(points)=-U(points-1);
    BU(2:points)=U(2:points)-U(1:points-1);
    BU(1)=U(1);

    AP(1:points-1)=P(2:points)-P(1:points-1);
    AP(points)=0;
    BP(2:points)=P(2:points)-P(1:points-1);
    BP(1)=0;

    coffpos=inv(matA)*[AP;AU];
    coffnec=inv(matA)*[BP;BU];


    P(1:100)=P(1:100)-dtdx*(coffnec(2,1:100) + coffpos(1,1:100));
    U(1:100)=U(1:100)-dtdx*(coffnec(2,1:100) - coffpos(1,1:100));

    P(101:200)=P(101:200)-dtdx*(coffnec(2,101:200) + coffpos(1,101:200))*0.5;
    U(101:200)=U(101:200)-dtdx*(coffnec(2,101:200) - coffpos(1,101:200))*0.5;

end
```

Q2 second Order System @t=6

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-5,5]; % interval
points=200; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
U=zeros(points,1)';P=zeros(points,1)';
U(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi)); % intial conditions
P(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi));
Ui=U;
Pi=P;
tim=6;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
matA=[-1 1;1 1];
for i=1:n
    % A be forward difference and B to be backward
    AU(1:points-1)=U(2:points)-U(1:points-1);
    AU(points)=-U(points-1);
    BU(2:points)=U(2:points)-U(1:points-1);
    BU(1)=U(1);

    AP(1:points-1)=P(2:points)-P(1:points-1);
    AP(points)=0;
    BP(2:points)=P(2:points)-P(1:points-1);
    BP(1)=0;

    coffpos=inv(matA)*[AP;AU];
    coffnec=inv(matA)*[BP;BU];
```

```
    mupos=(1+dtdx)/2;
    munev=(1-dtdx)/2;

    PFnev(1:100)=mupos*coffnec(2,1:100) + munev*coffnec(1,1:100);
    PFpos(1:100)=munev*coffpos(2,1:100) + mupos*coffpos(1,1:100);

    UFnev(1:100)=mupos*coffnec(2,1:100) - munev*coffnec(1,1:100);
    UFpos(1:100)=munev*coffpos(2,1:100) - mupos*coffpos(1,1:100);

    mupos=(1+0.5*dtdx)/2;
    munev=(1-0.5*dtdx)/2;

    PFnev(101:200)=(mupos*coffnec(2,101:200) + munev*coffnec(1,101:200))*0.5;
    PFpos(101:200)=(munev*coffpos(2,101:200) + mupos*coffpos(1,101:200))*0.5;

    UFnev(101:200)=(mupos*coffnec(2,101:200) - munev*coffnec(1,101:200))*0.5;
    UFpos(101:200)=(munev*coffpos(2,101:200) - mupos*coffpos(1,101:200))*0.5;

    P=P-dtdx*(PFnev + PFpos);
    U=U-dtdx*(UFnev + UFpos);


end
```

Q3 First Order @t=6

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-5,5]; % interval
points=200; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
U=zeros(points,1)';P=zeros(points,1)';
U(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi)); % intial conditions
P(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi));
Ui=U;
Pi=P;
tim=6;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
matA=[-1 1;1 1];
AU=[];BU=[];AP=[];BP=[];
for i=1:n
    % A be forward difference and B to be backward
    AU(1:points-1)=U(2:points)-U(1:points-1);
    AU(points)=-U(points-1);
    BU(2:points)=U(2:points)-U(1:points-1);
    BU(1)=U(1);

    AP(1:points-1)=P(2:points)-P(1:points-1);
    AP(points)=0;
    BP(2:points)=P(2:points)-P(1:points-1);
    BP(1)=0;

    coffpos=inv(matA)*[AP;AU];
    coffnec=inv(matA)*[BP;BU];
```

```
    P(1:100)=P(1:100)-dtdx*(coffnec(2,1:100) + coffpos(1,1:100));
    U(1:100)=U(1:100)-dtdx*(coffnec(2,1:100) - coffpos(1,1:100));

    P(101:200)=P(101:200)-dtdx*(coffnec(2,101:200) + coffpos(1,101:200));
    U(101:200)=U(101:200)-dtdx*(coffnec(2,101:200) - coffpos(1,101:200))*0.5;

end
```

Q3 second Order @t=6

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-5,5]; % interval
points=200; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
U=zeros(points,1)';P=zeros(points,1)';
U(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi)); % intial conditions
P(21:60)=0.5*(1 + cos((dom(21:60) + 3)*pi));
Ui=U;
Pi=P;
tim=6;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
matA=[-1 1;1 1];
for i=1:n
    % A be forward difference and B to be backward
    AU(1:points-1)=U(2:points)-U(1:points-1);
    AU(points)=-U(points-1);
    BU(2:points)=U(2:points)-U(1:points-1);
    BU(1)=U(1);

    AP(1:points-1)=P(2:points)-P(1:points-1);
    AP(points)=0;
    BP(2:points)=P(2:points)-P(1:points-1);
    BP(1)=0;

    coffpos=inv(matA)*[AP;AU];
    coffnec=inv(matA)*[BP;BU];

    mupos=(1+dtdx)/2;
    munev=(1-dtdx)/2;

    PFnev(1:100)=mupos*coffnec(2,1:100) + munev*coffnec(1,1:100);
    PFpos(1:100)=munev*coffpos(2,1:100) + mupos*coffpos(1,1:100);

    UFnev(1:100)=mupos*coffnec(2,1:100) - munev*coffnec(1,1:100);
    UFpos(1:100)=munev*coffpos(2,1:100) - mupos*coffpos(1,1:100);

    mupos=(1+0.5*dtdx)/2;
    munev=(1-0.5*dtdx)/2;

    PFnev(101:200)=(mupos*coffnec(2,101:200) + munev*coffnec(1,101:200));
    PFpos(101:200)=(munev*coffpos(2,101:200) + mupos*coffpos(1,101:200));

    UFnev(101:200)=(mupos*coffnec(2,101:200) - munev*coffnec(1,101:200))*0.5;
```

```
    UFpos(101:200)=(munev*coffpos(2,101:200) - mupos*coffpos(1,101:200))*0.5;

    P=P-dtdx*(PFnev + PFpos);
    U=U-dtdx*(UFnev + UFpos);


end
```

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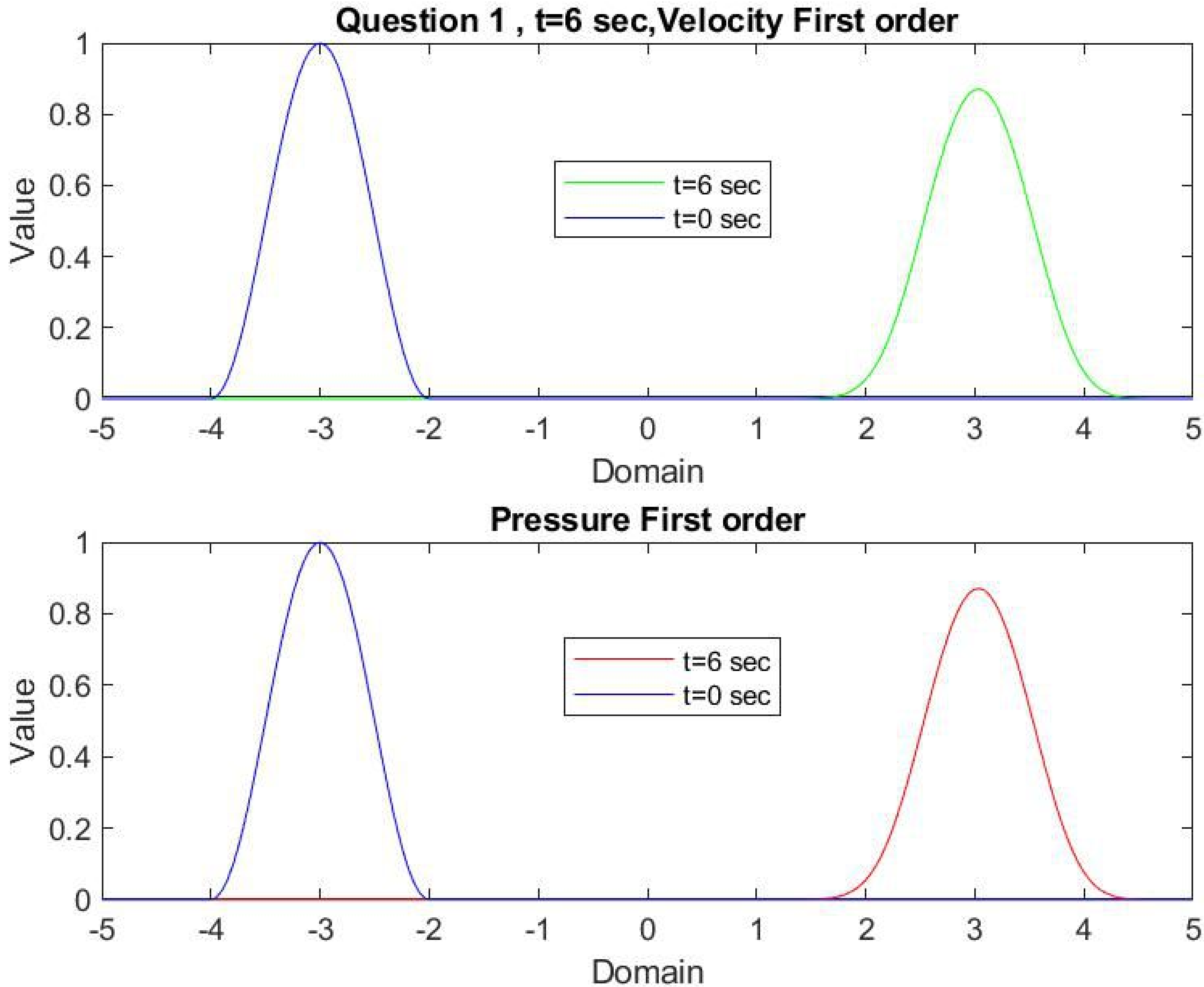
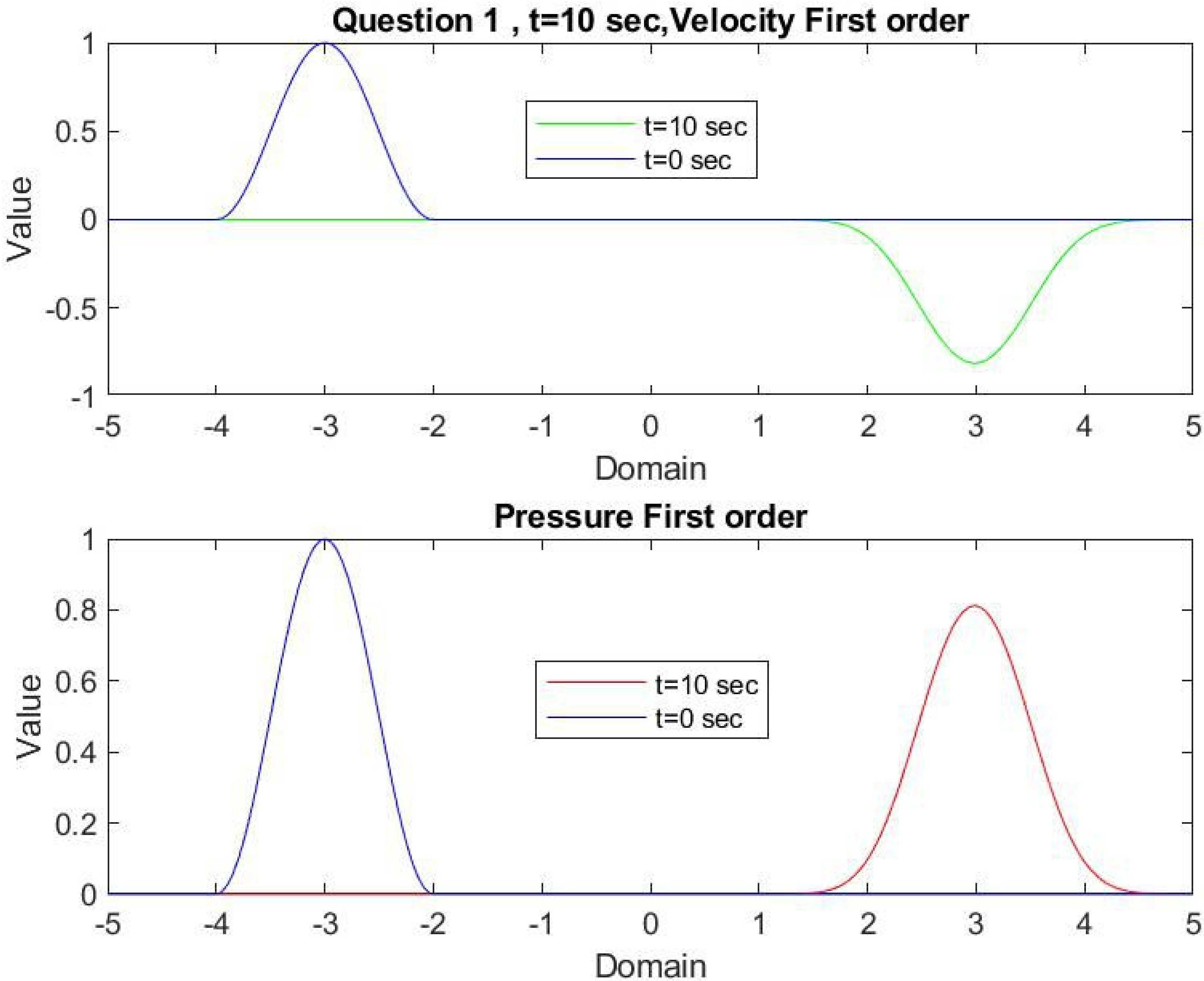
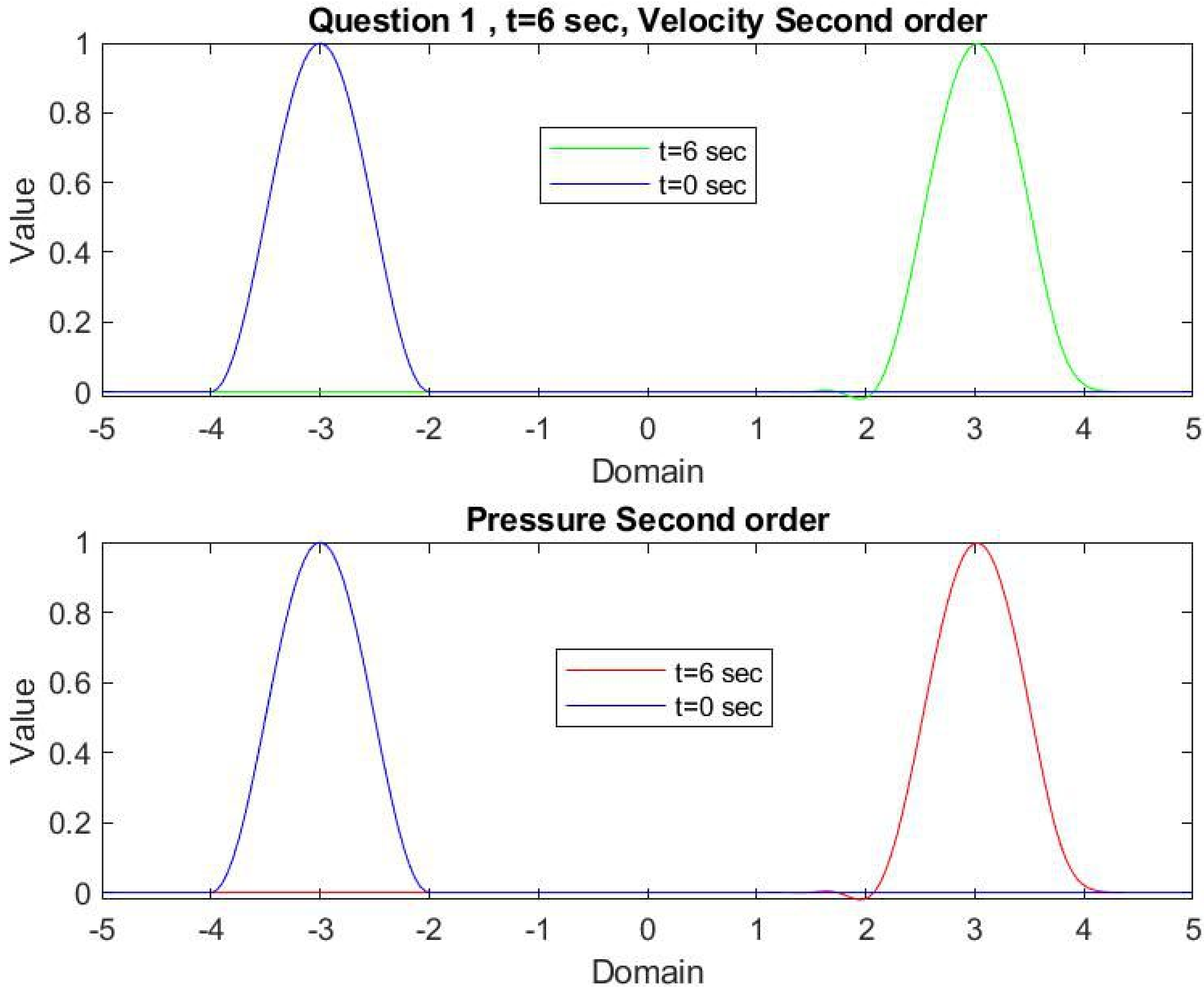


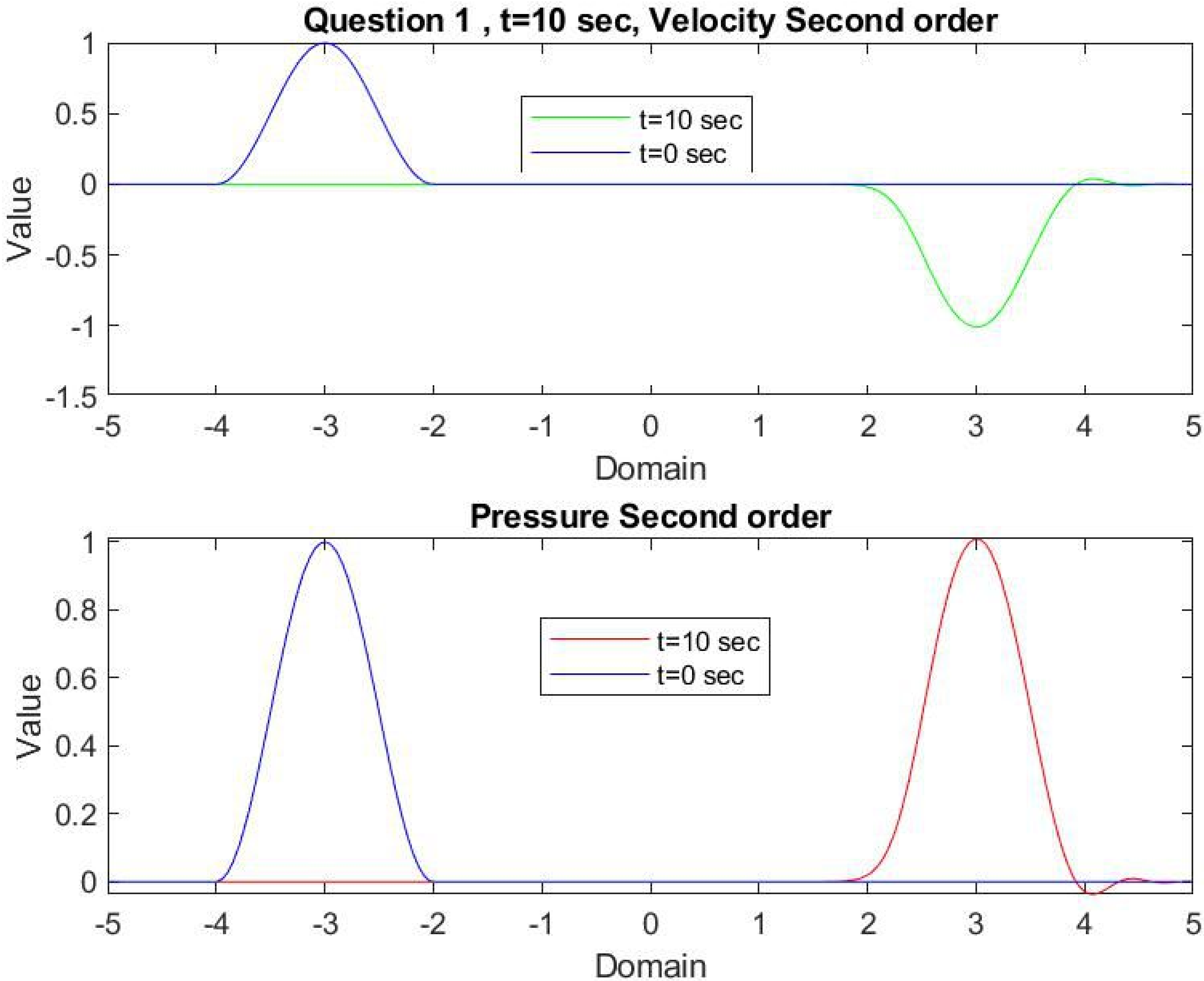
Question 1 , t=6 sec,Velocity First order
t=6 sec
t=0 sec
Value
Domain
Pressure First order
t=6 sec
t=0 sec
Value
Domain

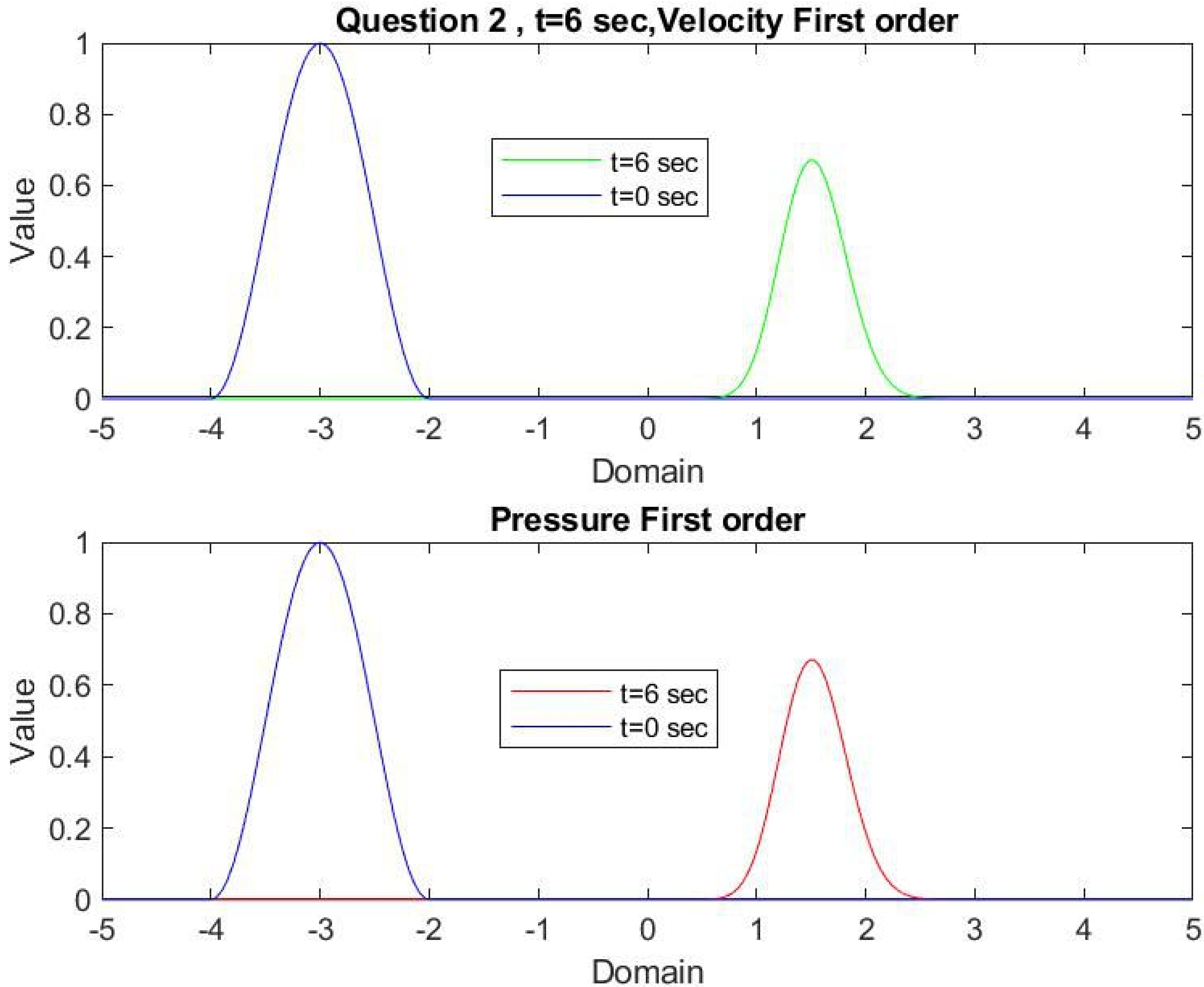


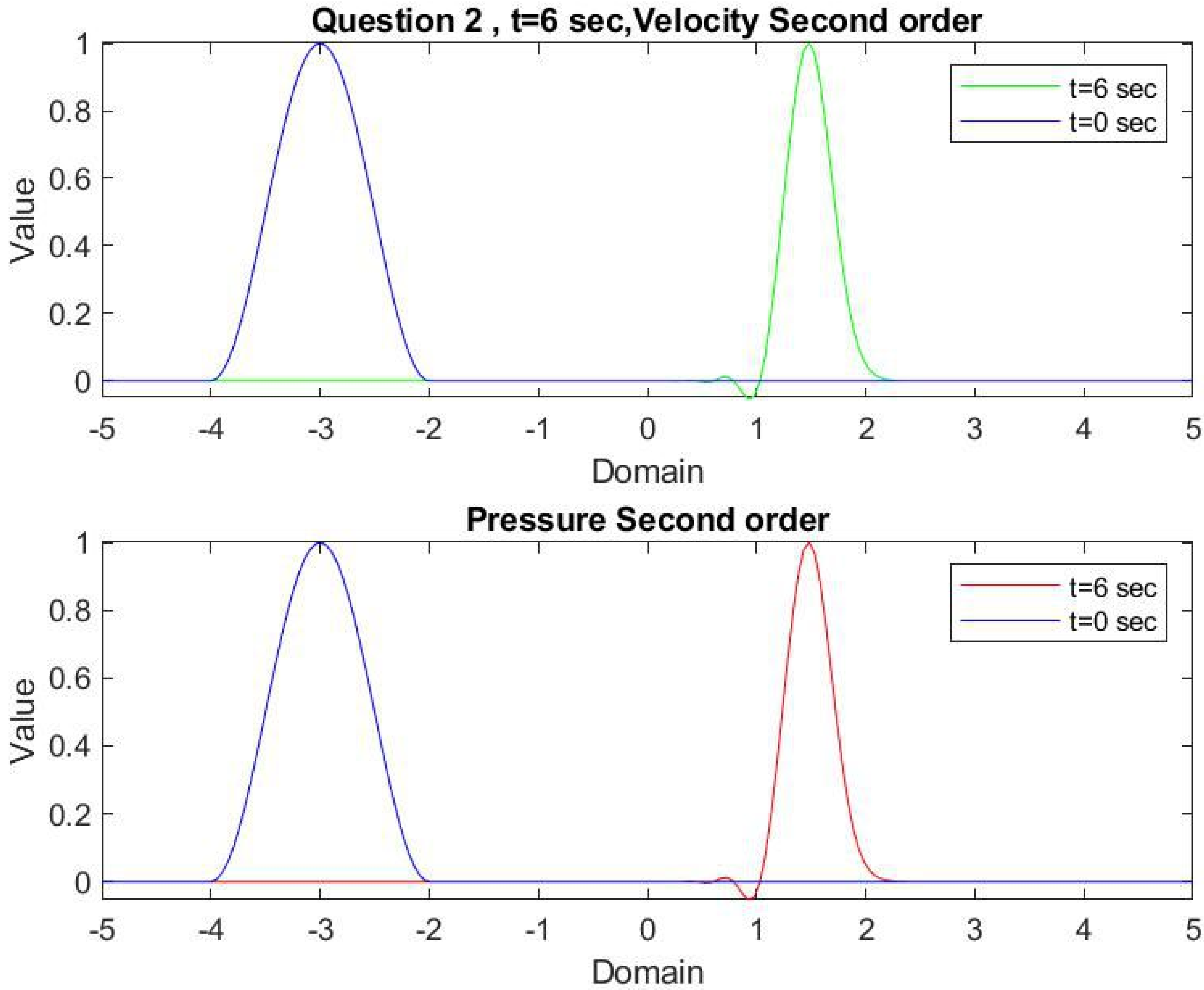
Question 1 , t=10 sec,Velocity First order
t=10 sec
t=0 sec
Value
Domain
Pressure First order
t=10 sec
t=0 sec
Value
Domain



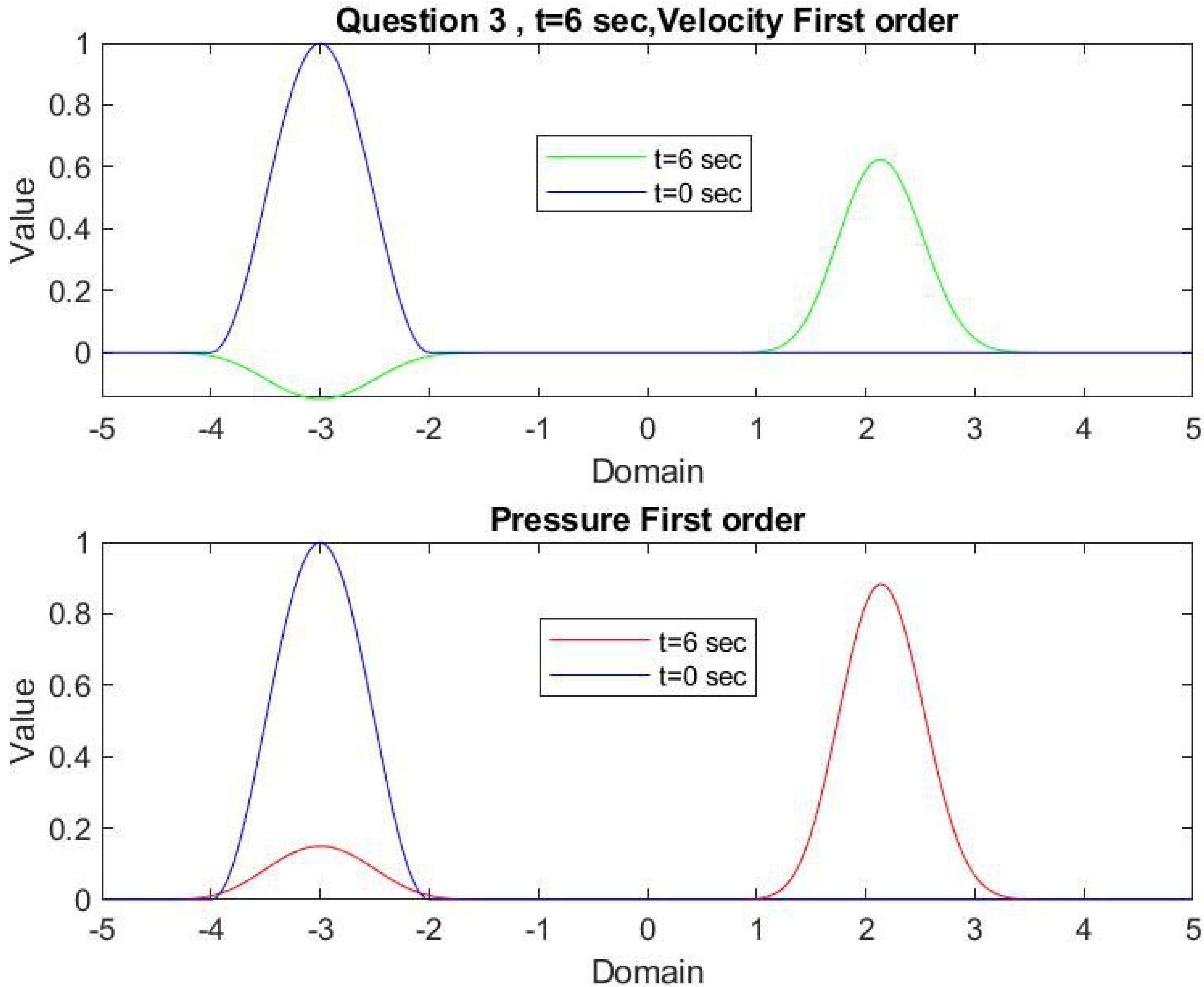
Question 1 , t=6 sec, Velocity Second order
t=6 sec
t=0 sec
Value
Domain
Pressure Second order
t=6 sec
t=0 sec
Value
Domain

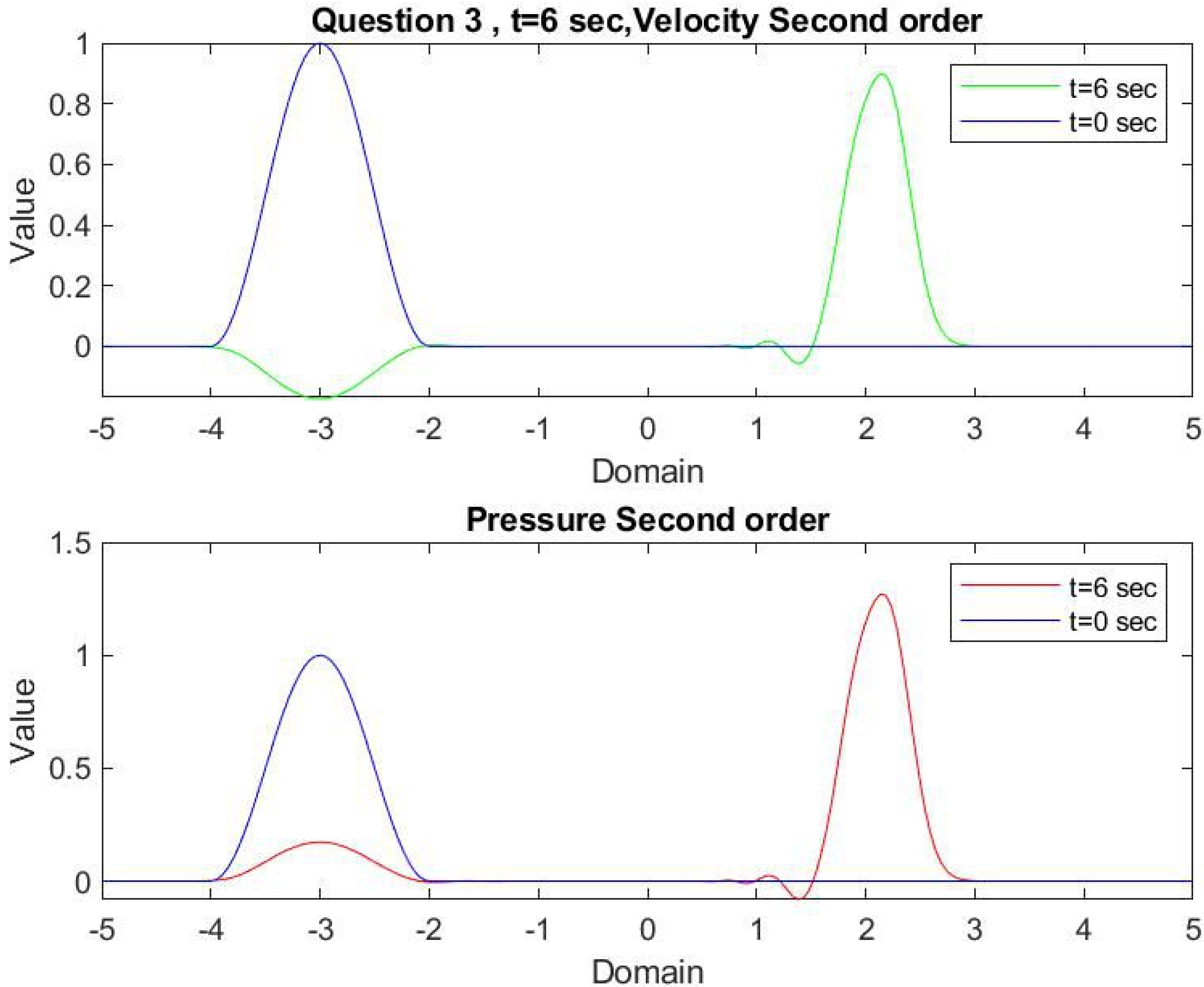

Question 1 , t=10 sec, Velocity Second order
t=10 sec
t=0 sec
Value
Domain
1
0.5
0
-0.5
-1
-1.5
-5
-4
-3
-2
-1
0
1
2
3
4
5
Pressure Second order
t=10 sec
t=0 sec
Value
Domain
1
0.8
0.6
0.4
0.2
0


Question 2 , t=6 sec,Velocity First order
Value
Domain
t=6 sec
t=0 sec
Pressure First order
Value
Domain
t=6 sec
t=0 sec



Question 2 , t=6 sec,Velocity Second order
t=6 sec
t=0 sec
Value
Domain
Pressure Second order
t=6 sec
t=0 sec
Value
Domain


Question 3 , t=6 sec,Velocity First order
Value
Domain
t=6 sec
t=0 sec
-5
-4
-3
-2
-1
0
1
2
3
4
5
0
0.2
0.4
0.6
0.8
1
Pressure First order
Value
Domain
t=6 sec
t=0 sec


Question 3 , t=6 sec,Velocity Second order
t=6 sec
t=0 sec
Value
Domain
Pressure Second order
t=6 sec
t=0 sec
Value
Domain